AI Impact on Job Sectors: Data Analysis and Visualization

1. Project Overview and Objective

This project focuses on analysing the impact of Artificial Intelligence (AI) on different job sectors using a dataset related to employment, salary trends, automation risk, upskilling requirements, productivity, and job satisfaction. The raw data was cleaned and transformed using Microsoft Excel to improve data quality and prepare it for analysis. An interactive Power BI dashboard was then created to visualize the data and derive meaningful insights.

The main objective of this project is to analyse the impact of AI on employment, job opportunities, and salary trends across different job sectors using Excel and Power BI dashboards. The project also aims to understand workforce characteristics, automation risk levels, upskilling requirements, productivity changes, and job satisfaction to identify how AI is influencing different aspects of the job market.

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2. Data Sources

- **Source Description and Timeline:** The dataset used for this project was obtained from **Kaggle**. It contains employee and job-related data used to analyse the impact of Artificial Intelligence (AI) across different job sectors. The dataset includes information related to employee demographics, industry, job roles, work experience, AI adoption levels, automation risk, upskilling requirements, salary before and after AI adoption, job status, productivity changes, remote work, and job satisfaction.No specific time period or year range is explicitly defined in the dataset. Therefore, the analysis focuses on the available employee and job sector data without restricting it to a particular timeline.
- **Domain**: Artificial Intelligence (AI), Employment and Workforce Analytics / Job Market Analysis

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3. Problem Statement

- To analyse the impact of Artificial Intelligence (AI) on employment and job opportunities across different job sectors.

- To compare salary trends before and after AI adoption across industries and understand the financial impact of AI on employees.
- To identify employees and job sectors with different levels of automation risk and analyse the need for upskilling.
- To examine the relationship between AI adoption, productivity changes, job satisfaction, and job status to understand the overall impact of AI on the workforce.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Employee_ID	Integer / String	Unique identifier for each employee
Age	Numeric (Integer)	Age of the employee in years
Gender	Categorical	Gender of the employee
Industry	Categorical	Industry in which the employee works
Job_Role	Categorical	Job position or role of the employee
Years_Experience	Numeric(Integer)	Total years of professional experience
AI_Adoption_Level	Categorical	Level of AI adoption in the employee's industry or workplace
Automation_Risk	Categorical	Risk level of the employee's job being automated by AI
Upskilling_Required	Categorical	Indicates whether additional skills or training are required
Salary_Before_AI	Numeric	Employee's salary before AI implementation
Salary_After_AI	Numeric	Employee's salary after AI implementation

Job_Status	Categorical	Indicates whether the employee's job was Modified, Replaced, or Unchanged
Work_Hours_Per_Week	Numeric	Number of hours worked by the employee per week
Remote_Work	Categorical	Indicates whether the employee works remotely
Job_Satisfaction	Numeric	Employee's job satisfaction score
Productivity_Change_%	Numeric	Percentage change in employee productivity after AI adoption

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5. Tools & Technologies

- **Microsoft Excel:** Used for data cleaning, handling missing values, removing duplicates, data transformation, formatting, and creating Pivot Tables for data analysis.
- **Microsoft Power BI:** Used for data modelling, creating relationships between tables, writing DAX measures, performing data analysis, designing visualizations, and developing interactive dashboards.

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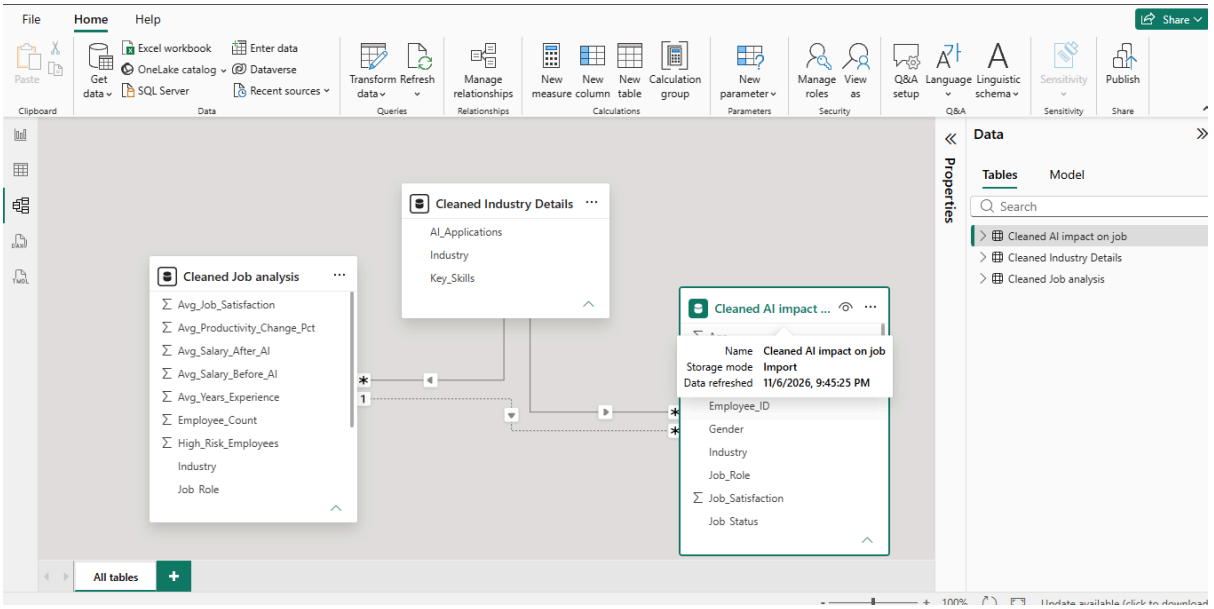
6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Identified and removed duplicate records, handled missing values, corrected inconsistent entries, and standardized data formats.
- **Data Validation:** Checked employee details, job roles and industry information for consistency and accuracy.
- **Filtering & Sorting:** Applied filters and sorting to organize employee records and focus on relevant data for analysis.
- **Data Formatting:** Standardized column names, corrected data types, and formatted numerical and categorical fields.
- **Calculated Fields:** Prepared relevant fields for salary comparison, productivity analysis, and AI impact assessment.
- **Pivot Tables:** Created Pivot Tables to summarize employee information, average salaries, automation risks, and industry-wise analysis.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables using common key columns and defined appropriate cardinality between related tables.



- **Calculated Columns & DAX Measures:** Implemented DAX formulas to calculate important performance indicators and analyze the impact of AI on employees, salaries, productivity, and automation risks.

- **List of DAX Measures:**

Measure Name	Purpose
Total Employees	Calculates the total number of employees
Average Age	Calculates the average age of employees
Average Salary Before AI	Calculates average salary before AI adoption
Average Salary After AI	Calculates average salary after AI adoption
Salary Increase %	Calculates percentage change in average salary
Average Salary by Industry	Calculates average salary for each industry

Highest Salary Industry	Identifies the industry with the highest average salary
High Risk Employees	Counts employees with High automation risk
Medium Risk Employees	Counts employees with Medium automation risk
Low Risk Employees	Counts employees with Low automation risk
Average Productivity Change	Calculates average productivity change after AI adoption
Average Job Satisfaction	Calculates average employee job satisfaction

3. Key DAX Functions Used:

- COUNTROWS()
- DISTINCTCOUNT()
- AVERAGE()
- CALCULATE()
- FILTER()
- DIVIDE()
- MAX()
- SUM()

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8. Analysis and Visualizations (Power BI)

Dashboard Features: Created interactive Power BI dashboards to analyze the impact of Artificial Intelligence on employment, salary trends, productivity, and automation risks across different job sectors.

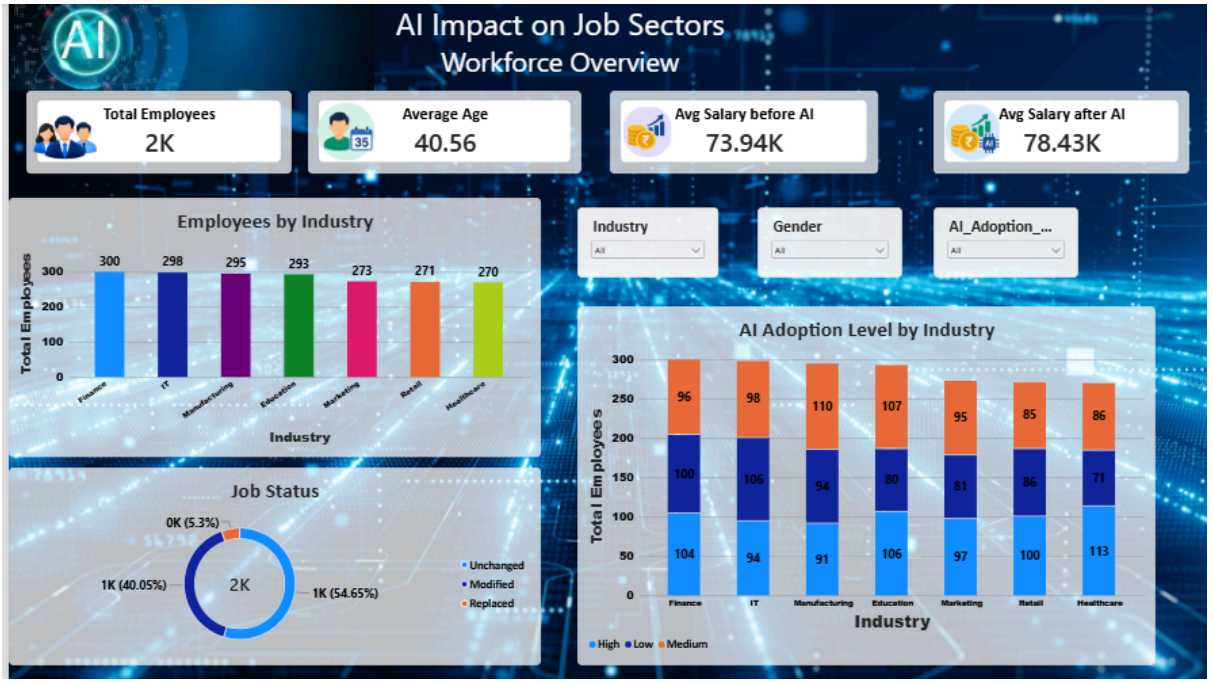
List of Visualizations Used:

Visualization	**Purpose**
Card Visuals	Display Total Employees, Average Age, Average Salary Before AI and After AI
Column Chart	Analyze employee distribution across industries

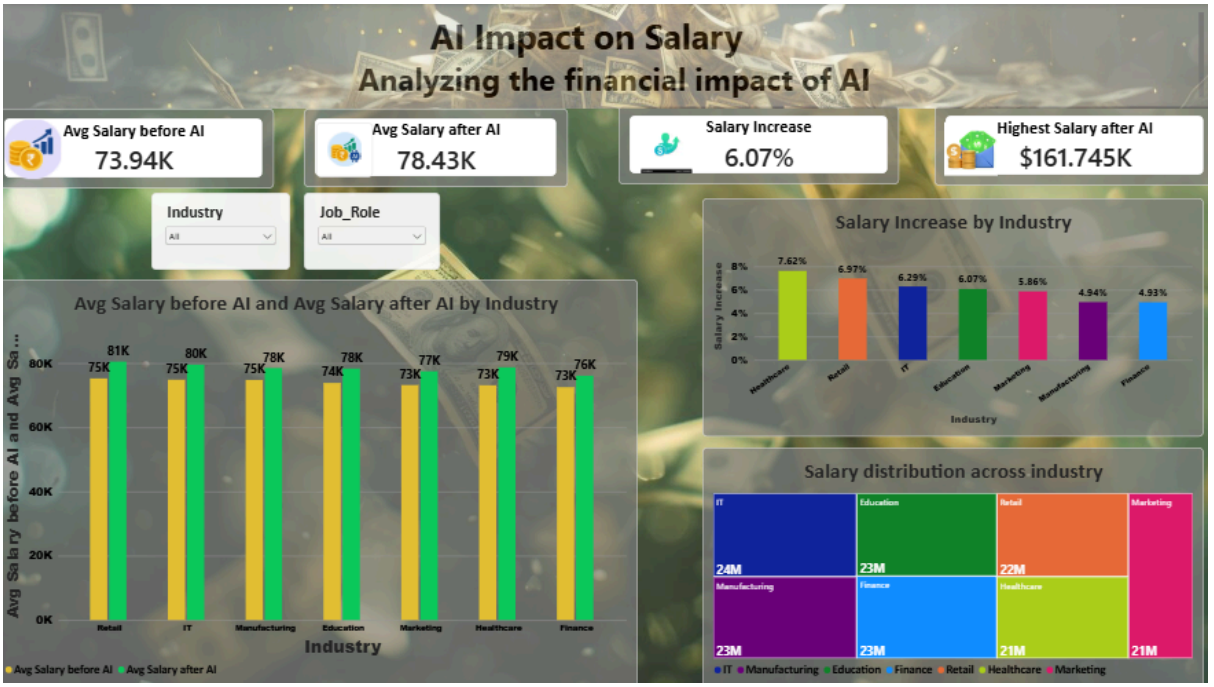
Stacked Column Chart	Analyze AI Adoption Level across different industries
Clustered Column Chart	Compare salary trends before and after AI adoption
Column Chart	Analyse salary increase by Industry
TreeMap	Compare salary distribution across industries
Stacked Bar Chart	Analyse automation risk level distribution across industries
Funnel Chart	Compare Upskilling required across industry
Table	Key skills required based on industries
Scatter Chart	Compare Productivity Change and Job Satisfaction based on Job Role

Filters and Slicers: Added Industry, Gender, Job Role and AI Adoption Level slicers.

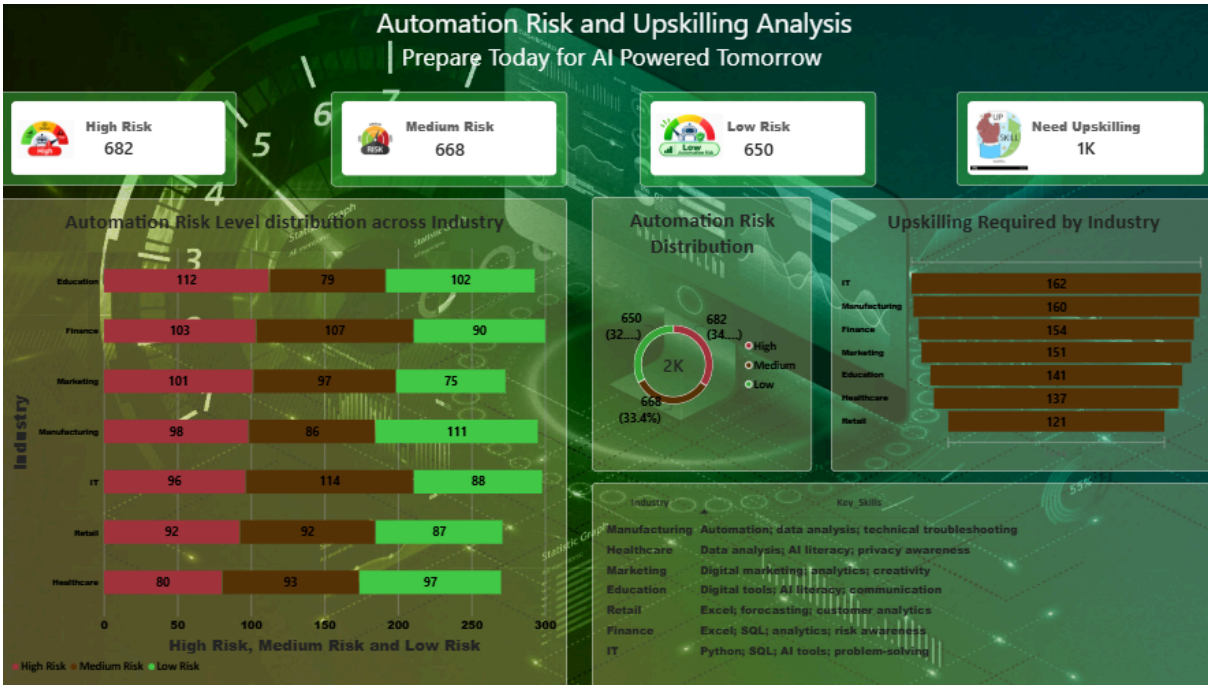
DASHBOARD 1: AI Impact on Job Sectors: Workforce Overview



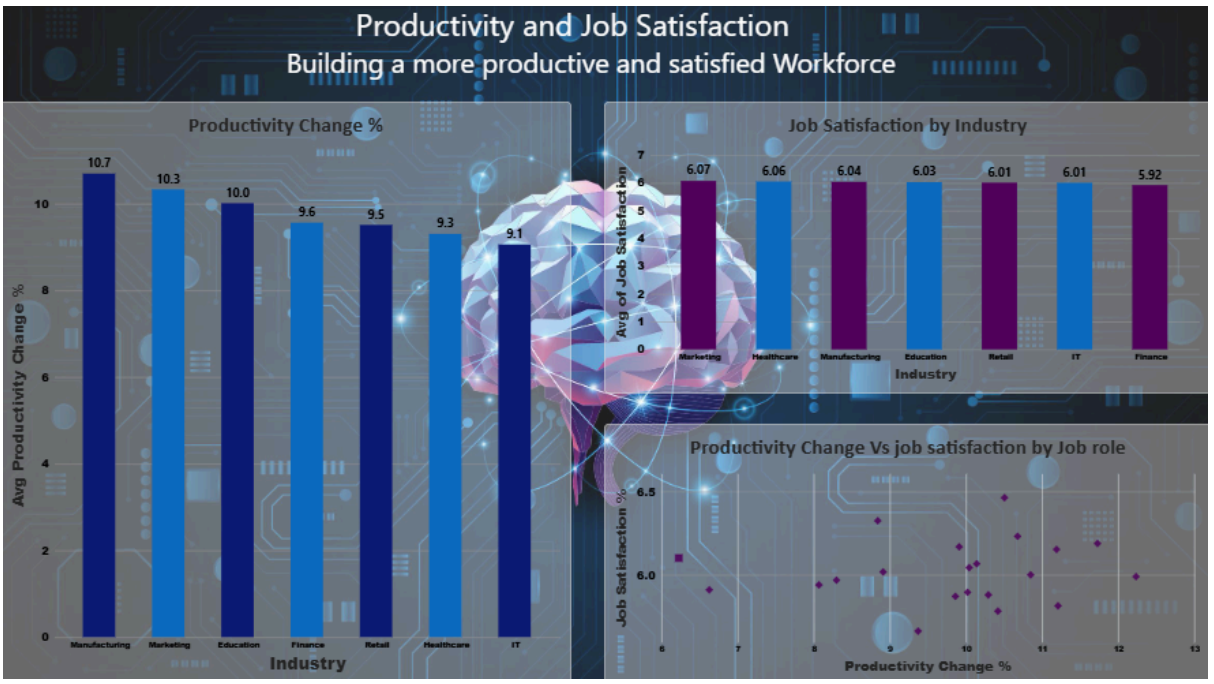
DASHBOARD 2: AI Impact On Salary



DASHBOARD 3 - Automation Risk and Upskilling Analysis:



DASHBOARD 4 - Productivity and Job Satisfaction



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9. Insights & Conclusions

Key Findings:

- Identified the impact of AI adoption on employee salaries across different industries.
- Analyzed the distribution of employees across High, Medium, and Low automation risk levels.
- Compared average salaries before and after AI adoption to understand salary changes.
- Identified industries with higher average salaries.
- Examined changes in employee productivity after AI implementation.
- Analyzed the relationship between AI adoption, job status, and employment opportunities.
- Evaluated employee job satisfaction across different job roles and industries.

Analysis Insights:

A. Descriptive Analytics – What happened?

- Analyzed the total number of employees across different industries and job roles.
- Calculated average employee age and average salaries before and after AI adoption.

- Identified the distribution of automation risk levels.
- Summarized employee job status and productivity changes.

B. Diagnostic Analytics – Why did it happen?

- Compared salary differences across industries to understand variations after AI adoption.
- Examined how AI adoption levels are associated with automation risk.
- Explored the relationship between productivity changes and job satisfaction.

C. Predictive Analytics – What might happen in the future?

- Identified job roles that may face higher automation risks based on current data patterns.
- Evaluated industries that may experience greater AI-driven workforce changes.
- Identified potential changes in employment opportunities due to increasing AI adoption.
- Highlighted the importance of future skill development and upskilling.

D. Prescriptive Analytics – What should be done?

- Encourage employees in high-risk job roles to develop AI-related and technical skills.
- Recommend upskilling and reskilling programs to improve employability.
- Encourage organizations to adopt AI while supporting affected employees.
- Promote continuous learning to adapt to changing job requirements.
- Focus on combining human expertise with AI technologies to improve productivity.

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10. Conclusions

The project analyzes the impact of Artificial Intelligence on employment, salary trends, productivity, and automation risks across different industries. Using Microsoft Excel and Power BI, the dataset was cleaned, transformed, analyzed, and visualized through interactive dashboards.

The analysis helps identify employment patterns, understand potential automation risks, evaluate salary changes, and highlight the importance of upskilling and reskilling in an AI-driven workplace.